

Assignment 5 – Interactive Todo List

StatefulWidget & setState State Management • Add, Delete, & Mark-Complete

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1. Assignment Overview

This assignment requires building a fully functional, interactive **Todo List** in Flutter by mastering `StatefulWidget` and `setState()`. The app provides a reactive user experience supporting adding tasks with modal bottom sheet validation, marking tasks as completed with strikethrough styling, deleting tasks via swipe or button with undo support, priority levels, categories, and real-time completion progress tracking.

ADD Task Creation

Opens a modal bottom sheet with required title validation, optional description, priority chips (Low, Med, High), and category tags. Inserts into the state list via `setState()`.

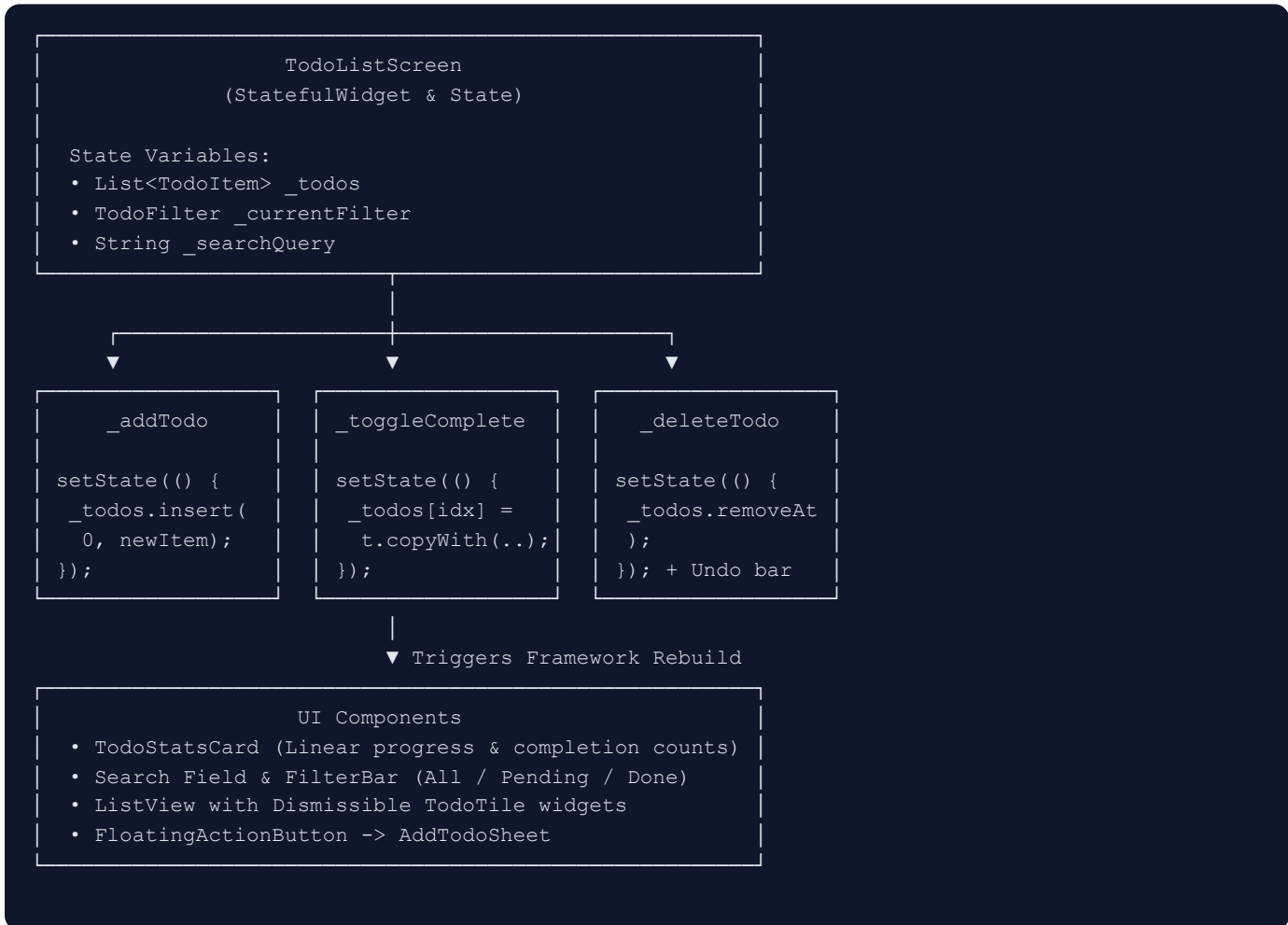
TOGGLE Mark-Complete

Tapping the custom checkbox mutates the task's `isCompleted` flag via `copyWith()`, animating strike-through styling and updating the progress bar.

DELETE Swipe & Undo

Uses `Dismissible` for swipe-to-delete and explicit icon buttons. Triggers a floating `SnackBar` with an `UNDO` action to restore deleted tasks.

2. Architecture & State Flow



3. Key Implementation Code

A. Adding a Task via `setState()` :

```
void _addTodo(TodoItem item) {
  setState(() {
    _todos.insert(0, item);
  });
  ScaffoldMessenger.of(context).showSnackBar(
    SnackBar(content: Text('Added task "${item.title}")),
  );
}
```

B. Toggling Completion with Immutable `copyWith()` :

```
void _toggleTodoComplete(String id, bool? isCompleted) {
  final index = _todos.indexWhere((t) => t.id == id);
  if (index != -1) {
    setState(() {
      _todos[index] = _todos[index].copyWith(
        isCompleted: isCompleted ?? !_todos[index].isCompleted,
      );
    });
  }
}
```

C. Deleting with Undo Restoration:

```
void _deleteTodo(String id) {
  final index = _todos.indexWhere((t) => t.id == id);
  if (index == -1) return;
  final removed = _todos[index];
  setState(() {
    _todos.removeAt(index);
  });
  ScaffoldMessenger.of(context).showSnackBar(
    SnackBar(
      content: Text('Deleted "${removed.title}"),
      action: SnackBarAction(
        label: 'UNDO',
        onPressed: () {
          setState(() {
            _todos.insert(index, removed);
          });
        },
      ),
    ),
  );
}
```

4. Visual Verification & Output Screenshots

The application was run and validated in Google Chrome (`flutter run -d chrome`). The screenshots below demonstrate task addition, progress calculation, completion toggling, and tab filtering:

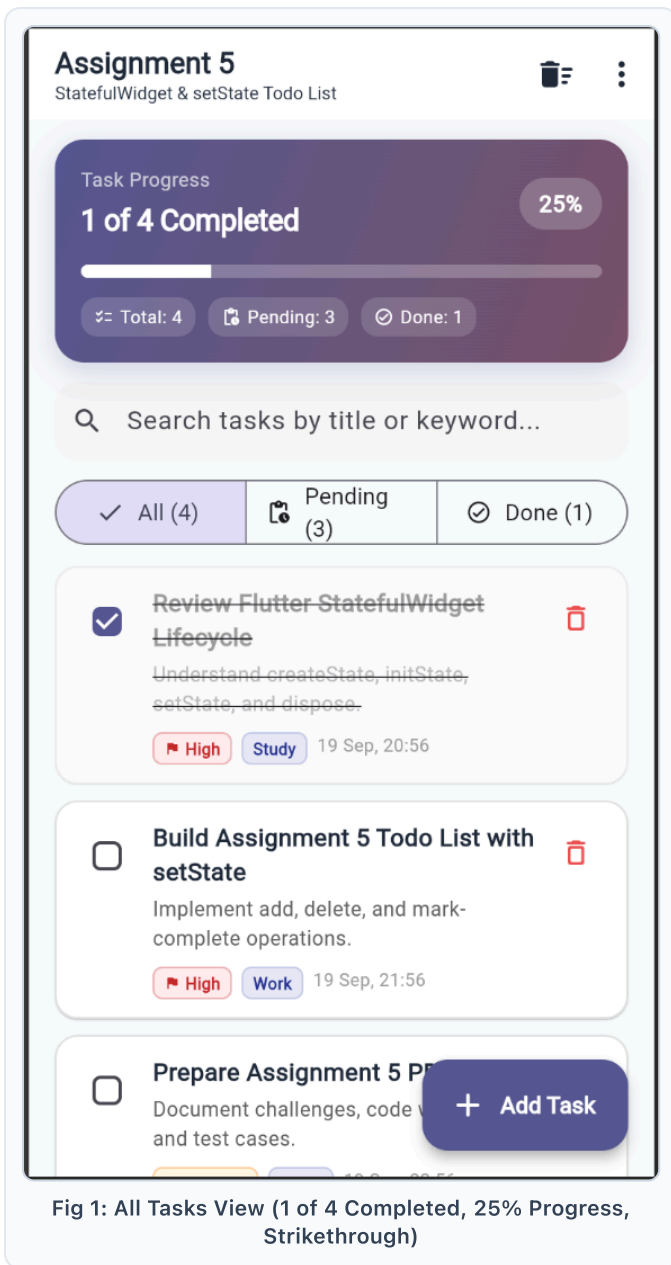


Fig 1: All Tasks View (1 of 4 Completed, 25% Progress, Strikethrough)

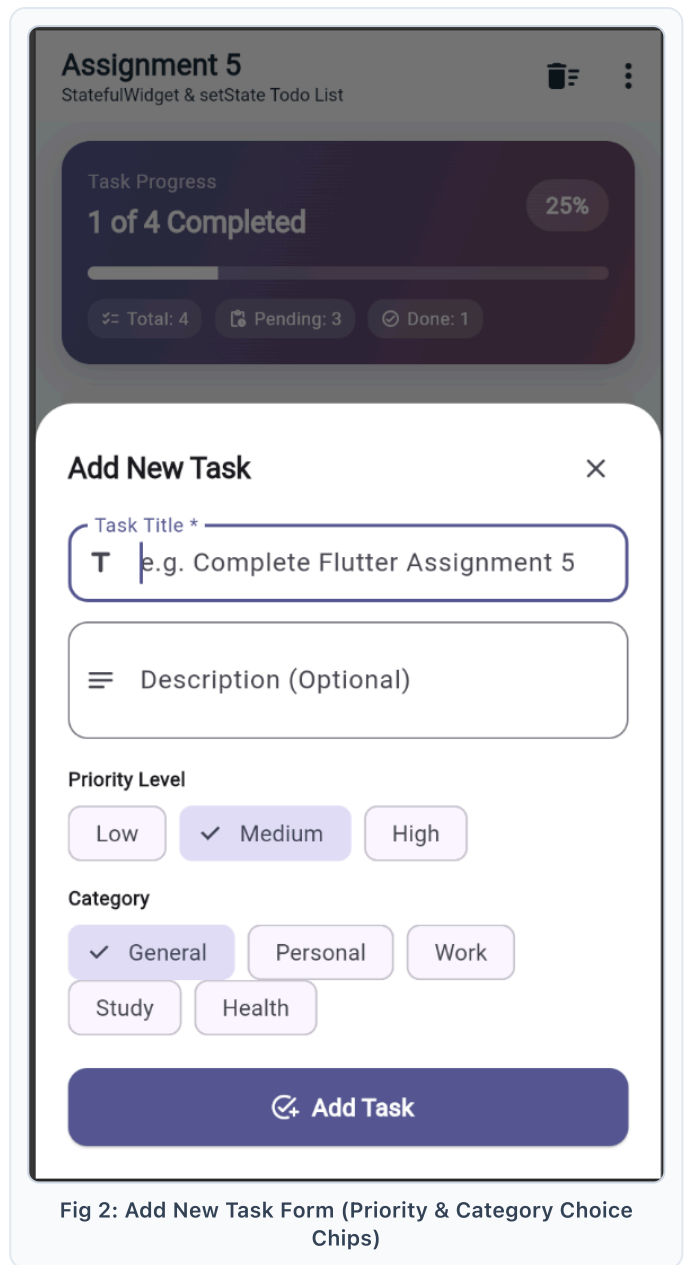
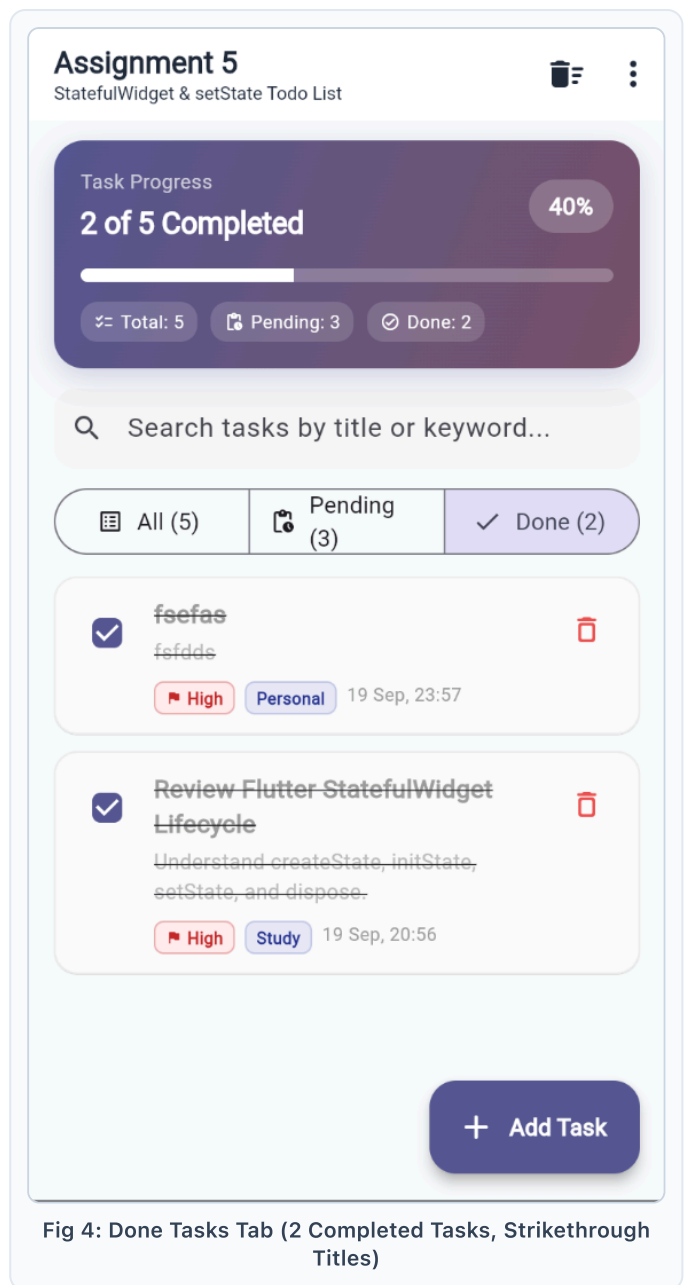
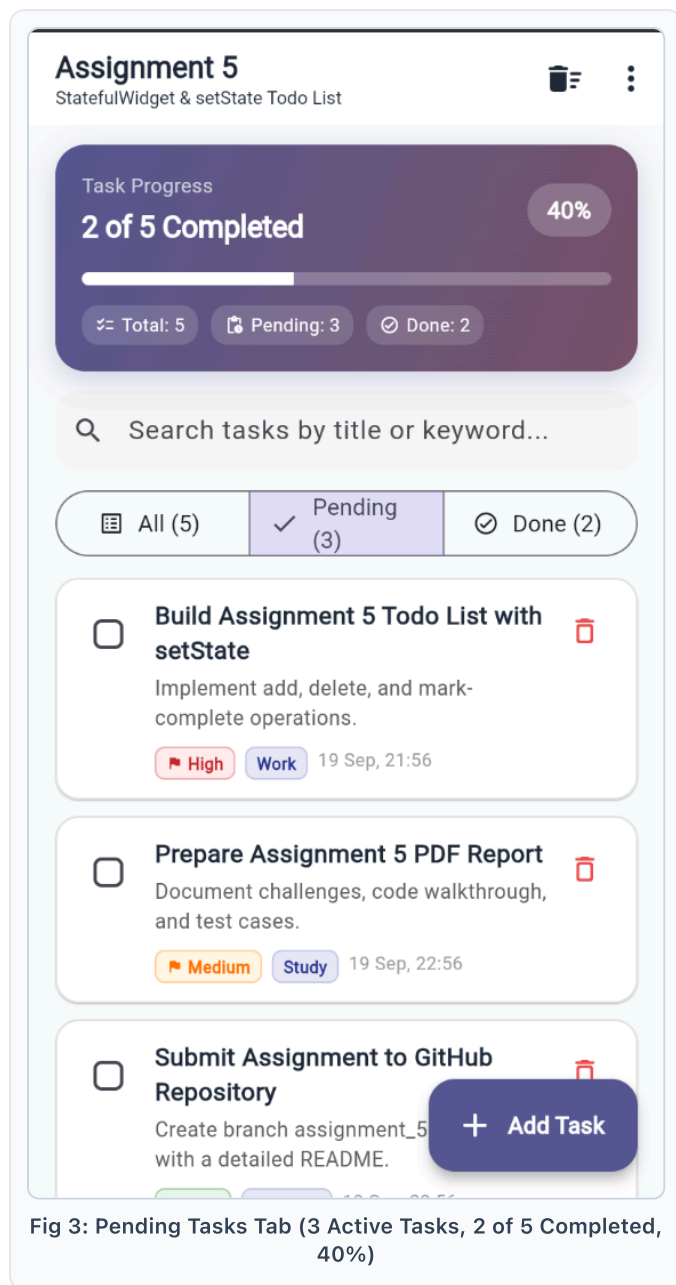


Fig 2: Add New Task Form (Priority & Category Choice Chips)



5. Challenges Faced & Solutions

Challenge	Technical Solution
Predictable State Mutation	Mutating list items in-place can cause subtle bugs where widgets don't redraw. Ensured <code>TodoItem</code> is immutable and utilized <code>copyWith()</code> inside <code>setState()</code> .
Dismissible Key Duplication	Using raw index keys in <code>Dismissible</code> widgets causes tree collision errors upon item deletion. Enforced globally unique keys (<code>ValueKey(todo.id)</code>).
Keyboard Insets in Bottom Sheet	The software keyboard can obscure submit buttons in modal sheets. Resolved by applying <code>MediaQuery.of(context).viewInsets.bottom</code> as bottom padding.
Restoring Original Index on Undo	Simply appending an undone item places it at the bottom. Preserved the exact removal index to re-insert the item cleanly with <code>_todos.insert(index, removed)</code> .

6. Automated Test Results

The solution includes **9 automated test cases** passing with 100% success:

Test File	Test Description	Result
todo_model_test.dart	Initializes with expected default values	PASS
todo_model_test.dart	copyWith updates fields while preserving unedited values	PASS
todo_model_test.dart	formattedDate produces formatted day, month, and time	PASS
todo_model_test.dart	Equality and hashCode verify value semantics	PASS
widget_test.dart	Renders initial tasks and progress summary accurately	PASS
widget_test.dart	Toggling task completion updates checkbox and completion stats	PASS
widget_test.dart	Adding a new task via AddTodoSheet inserts it into state	PASS
widget_test.dart	Deleting a task removes it from state with working undo action	PASS
widget_test.dart	Filter tabs switch between All, Pending, and Completed tasks	PASS

7. Static Analysis & Lints

Ran flutter analyze : 0 errors, 0 warnings, 0 lints. Adheres strictly to Flutter recommended practices and Material 3 design guidelines.